

MTH-645 Optimization Techniques

Assignment (1)

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It is required to find the minimum of the unimodal function $F(x, y) = 15e^{-x} + 2e^y$ along the segment joining the two points (1, 1) and (5, 4) within an interval of length 0.1. Use both Fibonacci search and Golden section methods to locate the minimum.

Solution:

a. Fibonacci search

$$L_0 = \sqrt{(5-1)^2 + (4-1)^2} = \sqrt{25} = 5.$$

$$L_n = 0.1.$$

$$reduction(r) = L_0/L_n = 5/0.1 = 50.$$

$$N_{fib} \geq r = 50 \implies N_{fib} = 55.$$

Fibonacci Sequence : 1, 1, 2, 3, 5, 8, 13, 21, 34, 55

$$iterations = 8.$$

The line segment between a and b is given by the vector:

$$L_0 = (5-1, 4-1) = (4, 3).$$

x_1 and x_2 are given by the equations:

$$x_1 = a + \frac{F_{n-2}}{F_n} L_0$$

$$x_2 = b - \frac{F_{n-2}}{F_n} L_0 = a + \frac{F_{n-1}}{F_n} L_0.$$

Iterations:

iteration	a	x_1	x_2	b	$f(x_1)$	comp	$f(x_2)$
1	(1,1)	(2.53, 2.15)	(3.47, 2.85)	(5,4)	18.36	<	35.04
2	(1,1)	(1.94, 1.70)	(2.53, 2.15)	(3.47, 2.85)	13.1	<	18.36
3	(1,1)	(1.59, 1.44)	(1.94, 1.70)	(2.53, 2.15)	11.5	<	13.1
4	(1,1)	(1.35, 1.26)	(1.59, 1.44)	(1.94, 1.70)	10.94	<	11.5
5	(1,1)	(1.24, 1.18)	(1.35, 1.26)	(1.59, 1.44)	10.85	<	10.94
6	(1,1)	(1.10, 1.08)	(1.24, 1.18)	(1.35, 1.26)	10.88	>	10.85
7	(1.10, 1.08)	(1.24, 1.18)	(1.21, 1.16)	(1.35, 1.26)	10.85	=	10.85

$$x^* = midpoint(a, b) = (1.23, 1.13)$$

b. Golden Section

First, let's compute the required number of iterations:

$$(0.618)^n = \frac{1}{50}$$

$$\implies n \ln(0.618) = \ln\left(\frac{1}{50}\right)$$

$$\implies n = 8.13$$

$$\implies n = 9.$$

x_1 and x_2 are given by the equations:

$$x_1 = a + 0.618L_0$$

$$x_2 = b - 0.618L_0$$

Iterations:

iteration	a	x_1	x_2	b	$f(x_1)$	comp	$f(x_2)$
1	(1.00, 1.00)	(2.52, 2.14)	(3.47, 2.85)	(5.00, 4.00)	18.29	<	35.18
2	(1.00, 1.00)	(1.94, 1.70)	(2.52, 2.14)	(3.47, 2.85)	13.18	<	18.29
3	(1.00, 1.00)	(1.58, 1.43)	(1.94, 1.70)	(2.52, 2.14)	11.50	<	13.18
4	(1.00, 1.00)	(1.36, 1.27)	(1.58, 1.43)	(1.94, 1.70)	10.97	<	11.50
5	(1.00, 1.00)	(1.22, 1.16)	(1.36, 1.27)	(1.58, 1.43)	10.84	<	10.97
6	(1.00, 1.00)	(1.13, 1.10)	(1.22, 1.16)	(1.36, 1.27)	10.83	<	10.84
7	(1.00, 1.00)	(1.08, 1.06)	(1.13, 1.10)	(1.22, 1.16)	10.86	>	10.83
8	(1.08, 1.06)	(1.13, 1.10)	(1.17, 1.12)	(1.22, 1.16)	10.83	=	10.83

$$x^* = \text{midpoint}(a, b) = (1.15, 1.11)$$

2.

It is required to find the maximum of the unimodal function $F = \ln x - 2x^2$ within the interval $(0, 1)$. Use the method which requires the minimum number of function evaluations to locate the maximum in an interval of length 0.05.

Solution:

We transform the problem into a minimization problem by multiply by -1 , we get: the new objective function:

$$G(x) = 2x^2 - \ln x$$

The **Fibonacci search** method is the method that requires the minimum number of function evaluations. So we use it for the solution.

$$L_0 = 1.$$

$$L_n = 0.05.$$

$$reduction(r) = L_0/L_n = 1/0.05 = 20.$$

$$N_{fib} \geq r = 20 \implies N_{fib} = 21.$$

Fibonacci Sequence : 1, 1, 2, 3, 5, 8, 13, 21

$$iterations = 6.$$

Iterations:

iteration	a	x_1	x_2	b	$f(x_1)$	comp	$f(x_2)$
1	0.00	0.38	0.61	1.00	1.25	>	1.24
2	0.38	0.61	0.76	1.00	1.24	<	1.43
3	0.38	0.52	0.61	0.76	1.19	<	1.24
4	0.38	0.47	0.52	0.61	1.19	>	1.19
5	0.47	0.52	0.57	0.61	1.19	<	1.21
6	0.47	0.50	0.53	0.56	1.19	<	1.20

$$x^* = midpoint(a, b) = 0.52$$

3.

- a. Use Fibonacci search to find the point of minimum energy $E = 14x^2 + y^2 + z^2 - 4xy - 6xz - 3x$ along the segment joining the two points $(1, 2, 4)$ and $(3, 5, 8)$ with an error $\epsilon \leq 0.05$.
- b. Comment on the relation between the gradient at the point of minimum and the segment direction.

Solution:

a.

$$\vec{L}_0 = (3 - 1, 5 - 2, 8 - 4) = (2, 3, 4).$$

$$L_0 = \sqrt{(3 - 1)^2 + (5 - 2)^2 + (8 - 4)^2} = 5.39.$$

$$L_n = 0.05.$$

$$\text{reduction}(r) = L_0/L_n = 5.39/0.05 = 107.8.$$

$$N_{fib} \geq r = 107.8 \implies N_{fib} = 144.$$

Fibonacci Sequence : 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144

$$\text{iterations} = 10.$$

Iterations:

iteration	a	x_1	x_2	b	$f(x_1)$	comp	$f(x_2)$
1	(1.00, 2.00, 4.00)	(1.76, 3.14, 5.52)	(2.23, 3.85, 6.47)	(3.00, 5.00, 8.00)	-1.97	<	-1.27
2	(1.00, 2.00, 4.00)	(1.47, 2.70, 4.94)	(1.76, 3.14, 5.52)	(2.23, 3.85, 6.47)	-1.91	>	-1.97
3	(1.47, 2.70, 4.94)	(1.76, 3.14, 5.52)	(1.94, 3.41, 5.88)	(2.23, 3.85, 6.47)	-1.97	<	-1.82
4	(1.47, 2.70, 4.94)	(1.65, 2.97, 5.30)	(1.76, 3.14, 5.52)	(1.94, 3.41, 5.88)	-1.99	<	-1.97
5	(1.47, 2.70, 4.94)	(1.58, 2.87, 5.16)	(1.65, 2.97, 5.30)	(1.76, 3.14, 5.52)	-1.98	>	-1.99
6	(1.58, 2.87, 5.16)	(1.65, 2.97, 5.30)	(1.69, 3.04, 5.39)	(1.76, 3.14, 5.52)	-1.99	<	-1.99
7	(1.58, 2.87, 5.16)	(1.62, 2.93, 5.25)	(1.65, 2.97, 5.30)	(1.69, 3.04, 5.39)	-1.99	>	-1.99
8	(1.62, 2.93, 5.25)	(1.65, 2.97, 5.30)	(1.66, 3.00, 5.33)	(1.69, 3.04, 5.39)	-1.99	>	-1.99
9	(1.65, 2.97, 5.30)	(1.66, 3.00, 5.33)	(1.67, 3.01, 5.35)	(1.69, 3.04, 5.39)	-1.99	<	-1.99
x	(1.65, 2.97, 5.30)	(1.66, 2.99, 5.32)	(1.66, 3.00, 5.33)	(1.67, 3.01, 5.35)	-1.99	>	-1.99

$$x^* = \text{midpoint}(a, b) = (1.66, 2.99, 5.33)$$

b.

$$\vec{\nabla}(E) = \left(\frac{\partial E}{\partial x}, \frac{\partial E}{\partial y}, \frac{\partial E}{\partial z} \right) = (28x - 4y - 6z - 3, 2y - 4x, 2z - 6x)$$

$$\nabla(E(1.66, 2.99, 5.33)) = (-0.46, -0.66, 0.7)$$

Let θ be the angle between the segment direction, $\vec{L}_0$, and the gradient direction at the point of minimum, $\vec{\nabla}(E(1.66, 2.99, 5.33))$. Then:

$$\begin{aligned} \cos(\theta) &= \frac{u \cdot v}{\|u\| \|v\|} \\ &= \frac{(2, 3, 4) \cdot (-0.46, -0.66, 0.7)}{5.39 * 3.02} \\ &= \frac{2 * (-0.46) + 3 * (-0.66) + 4 * 0.7}{5.39 * 1.07} = 0.03 \\ &\implies \theta = 88. \end{aligned}$$

This shows that the directions are nearly orthogonal.

4.

Consider the problem: Minimize $(x + 2y - 4)^2 + (3x - 4y - 2)^2$. Starting at $(0, 0)$, find the minimum along the steepest decent direction at $(0, 0)$, find the minimum along the steepest decent direction at $(0, 0)$ using Powell's quadratic interpolation and $\delta = 1$. How many quadratic interpolations are expected to lead to the minimum in this direction?

Solution:

Denote $F(x, y) = (x + 2y - 4)^2 + (3x - 4y - 2)^2$.

First, we compute the gradient:

$$\begin{aligned}\nabla(F(x, y)) &= \left(\frac{\partial E}{\partial x}, \frac{\partial E}{\partial y} \right) \\ \frac{\partial E}{\partial x} &= (2(x + 2y - 4)(1) + 2(3x - 4y - 2)(3)) \\ \frac{\partial E}{\partial y} &= 2(x + 2y - 4)(2) + 2(3x - 4y - 2)(-4) \\ \implies \nabla(F(0, 0)) &= (-20, 0)\end{aligned}$$

Next, we compute the steepest descent direction at $(0, 0)$:

$$\begin{aligned}-\nabla(F(0, 0)) &= (20, 0) \\ \vec{d} = \text{normalize}((20, 0)) &= (1, 0).\end{aligned}$$

Along the steepest descent direction, we parameterize (x, y) in terms of Δ :

$$(x, y) = (0, 0) + \Delta(1, 0) = (\Delta, 0)$$

And parametrize F in terms of Δ :

$$\begin{aligned}F(\Delta) &= F(\Delta, 0) = (\Delta - 4)^2 + (3\Delta - 2)^2 \\ &= 10\Delta^2 - 20\Delta + 20.\end{aligned}$$

Since only the initial guess point is given without a search interval, our strategy will be to take consecutive δ steps along the steepest descent direction until we find an appropriate search interval:

Δ	$F(\Delta)$
0	20
1	10
2	20

Since $F(\Delta) < F(0)$ and $F(\Delta) < F(2\Delta)$, then $[0, 2\Delta]$ is an appropriate search interval.

Next, we use Powell's quadratic interpolation at the 3 points: 0, Δ , 2Δ :

$$\begin{aligned}x_c &= \frac{a + b}{2} = \Delta \\ y &= x - x_c \\ G(y) &= ay^2 + by + c \\ G(0) &= c = F(\Delta) = 10 \\ G(\delta) &= G^+ = 20 \\ G(-\delta) &= G^- = 20\end{aligned}$$

(To simplify the calculation, I'll just look up the formula for y^* from the slides):

$$\begin{aligned}y^* &= -\frac{G^+ - G^-}{G^+ + G^- - 2G^0} \frac{\Delta}{2} \\ &= -\frac{(20 - 20)}{20 + 20 - 2 * 10} \frac{1}{2} = 0.\end{aligned}$$

$$\begin{aligned}x^* &= y^* + x_c \\ &= 0 + \Delta = 1.\end{aligned}$$

We don't need to perform further searches, since the current interval length $= 2\delta \implies \epsilon = \frac{2\delta}{2} = \delta$, achieves the required accuracy.

5.

Construct three simplices of the simplex method to find the minimum of $F = 5x^2 - 4xy + 10y^2 - 7x$ using the initial simplex with vertices: $(0,0)$, $(0.5, 0)$, $(0, 0.5)$. Show the steps on a graph paper.

Solution:

Assume an expansion coefficient $\gamma = 2$ and a contraction coefficient $\beta = 1/2$.

Iteration 1:

Step 1: Evaluate the function at all vertiex points:

$$\begin{array}{ll} x_1 = (0, 0) & F(x_1) = 0 \\ x_2 = (0.5, 0) & F(x_2) = -2.25 \\ x_3 = (0, 0.5) & F(x_3) = 2.5 \end{array}$$

$$\implies x_l = x_2, \quad x_h = x_3.$$

Step 2: Find the centroix point x_0 :

$$\begin{aligned} x_0 &= \frac{1}{2}(x_1 + x_2) = (0.25, 0) \\ F(x_0) &= -1.44. \end{aligned}$$

Step 3: Find the reflection point x_r :

$$\begin{aligned} x_r &= 2x_0 - x_h = (0.5, -0.5) \\ F(x_r) &= 1.25. \end{aligned}$$

Step 4: Decision point (accept OR expand OR contract):

$F(x_r) < F(x_h)$, but $F(x_r)$ is larger than all remaining points $\implies$ attemp contraction.

Step 5: Attempt contraction:

$$\begin{aligned} x_c &= \beta x_h + (1 - \beta)x_0 = (0.125, 0.25) \\ F(x_c) &= -0.3. \end{aligned}$$

$F(x_c) < \min[F(x_h), F(x_r)] \implies$ accept x_c .

Step 6: Test for convergence (standard-deviation):

$$Q = \sqrt{\frac{\sum_{i=1}^n [F(x_i) - F(x_0)]^2}{n}} = 1.16.$$

Iteration 2:

Step 1: Evaluate the function at all vertiex points:

$$\begin{array}{ll} x_1 = (0, 0) & F(x_1) = 0 \\ x_2 = (0.5, 0) & F(x_2) = -2.25 \\ x_3 = (0.125, 0.25) & F(x_3) = -0.3 \end{array}$$

$$\implies x_l = x_2, \quad x_h = x_1.$$

Step 2: Find the centroix point x_0 :

$$x_0 = \frac{1}{2}(x_2 + x_3) = (0.31, 0.125)$$

$$F(x_0) = -1.69.$$

Step 3: Find the reflection point x_r :

$$x_r = 2x_0 - x_h = (0.62, 0.25)$$

$$F(x_r) = -2.41$$

Step 4: Decision point (accept OR expand OR contract):

$F(x_r)$ is less than all other points $\implies$ attemp expansion.

Step 5: Attempt expansion:

$$x_e = 2x_r - x_0 = (0.93, 0.375)$$

$$F(x_e) = -2.17$$

$F(x_e) > F(x_r) \implies$ reject x_e and accept x_r .

Step 6: Test for convergence (standard-deviation):

$$Q = \sqrt{\frac{\sum_{i=1}^n [F(x_i) - F(x_0)]^2}{n}} = 0.96.$$

Iteration 3:

Step 1: Evaluate the function at all vertiex points:

$x_1 = (0.62, 0.25)$	$F(x_1) = -2.41$
$x_2 = (0.5, 0)$	$F(x_2) = -2.25$
$x_3 = (0.125, 0.25)$	$F(x_3) = -0.3$

$$\implies x_l = x_1, \quad x_h = x_3.$$

Step 2: Find the centroix point x_0 :

$$x_0 = \frac{1}{2}(x_1 + x_3) = (0.56, 0.125)$$

$$F(x_0) = -2.48.$$

Step 3: Find the reflection point x_r :

$$x_r = 2x_0 - x_h = (0.995, 0)$$

$$F(x_r) = -2.01$$

Step 4: Decision point (accept OR expand OR contract):

$F(x_r) < F(x_h)$, but $F(x_r)$ is larger than all remaining points $\implies$ attempt contraction.

Step 5: Attempt contraction:

$$\begin{aligned}x_c &= \beta x_h + (1 - \beta)x_0 = (0.34, 0.19) \\F(x_c) &= -1.7.\end{aligned}$$

$F(x_c) > F(x_r) \implies$ reject x_c and accept x_r .

Step 6: Test for convergence (standard-deviation):

$$Q = \sqrt{\frac{\sum_{i=1}^n [F(x_i) - F(x_0)]^2}{n}} = 0.3.$$

The final simplex is given by:

$$\begin{array}{ll}x_1 = (0.62, 0.25) & F(x_1) = -2.41 \\x_2 = (0.5, 0) & F(x_2) = -2.25 \\x_3 = (0.995, 0) & F(x_3) = -2.01.\end{array}$$

and the solution is given by:

$$\begin{aligned}x_0 &= \frac{1}{2}(x_1 + x_2) = (0.56, 0.125) \\F(x_0) &= -2.48.\end{aligned}$$

6.

Carry out three steps to find three new base points of pattern search to Minimize

$$F = x_1^2 x_2^2 + (x_1 - 2)^2 (x_2 - 3)^2$$

starting at $x = (1, 1)$, $\delta = 0.2$ for both directions. Show the steps taken on graph paper.

Solution:

Search directions (d_i):

$$\left((1, 0), (-1, 0), (0, 1), (0, -1) \right)$$

First search:

Step 1: Compute base point:

$$b^0 = (1, 1)$$

$$F(b^0) = 5.$$

Step 2: Make exploratory moves:

direction	b^0	$F(b^0)$	x^1	$F(x^1)$	$F(x^1)$ comp $f(b^0)$	ACCEPTED?
d_1	(1, 1)	5	(1.2, 1)	4	<	YES

Step 3: Make pattern move:

$$x^p = b^0 + 2\delta d_1 = (1.4, 1)$$

$$F(x^p) = 3.4.$$

direction	b^0	$F(b^0)$	x^1	$F(x^1)$	$F(x^1)$ comp $f(b^0)$	ACCEPTED?
d_1	(1, 1)	5	(1.6, 1)	3.2	<	YES

$$\Rightarrow b^1 = (1.6, 1).$$

Second search:

Step 1: Compute base point:

$$b^1 = (1.6, 1)$$

$$F(b^1) = 3.2.$$

Step 2: Make exploratory moves:

direction	b^1	$F(b^1)$	x^2	$F(x^2)$	$F(x^2)$ comp $f(b^1)$	ACCEPTED?
d_1	(1.6, 1)	3.4	(1.8, 1)	3.4	>	NO
d_2	(1.6, 1)	3.4	(1.4, 1)	3.4	>	NO
d_3	(1.6, 1)	3.4	(1.6, 1.2)	4.2	>	NO
d_4	(1.6, 1)	3.4	(1.6, 0.8)	2.4	<	YES

Step 3: Make pattern move:

$$x^p = b^1 + 2\delta d_4 = (1.6, 0.6)$$

$$F(x^p) = 1.8.$$

direction	b^1	$F(b^1)$	x^2	$F(x^2)$	$F(x^2)$ comp $f(b^1)$	ACCEPTED?
d_1	(1.6, 1)	5	(1.6, 0.4)	1.5	<	YES

$$\Rightarrow b^2 = (1.6, 0.4).$$

Third search:

Step 1: Compute base point:

$$b^2 = (1.6, 0.4)$$
$$F(b^2) = 1.5.$$

Step 2: Make exploratory moves:

direction	b^2	$F(b^2)$	x^3	$F(x^3)$	$F(x^3)$ comp $f(b^2)$	ACCEPTED?
d_1	(1.6, 0.4)	1.5	(1.8, 0.4)	0.79	<	ACCEPTED

Step 3: Make pattern move:

$$x^p = b^2 + 2\delta d_1 = (2, 0.4)$$
$$F(x^p) = 0.64.$$

direction	b^2	$F(b^2)$	x^3	$F(x^3)$	$F(x^3)$ comp $f(b^2)$	ACCEPTED?
d_1	(1.6, 0.4)	1.5	(2.2, 0.4)	1	<	YES

$$\implies b^3 = (2.2, 0.4).$$